① Randomization Unit and Analysis Unit 168.

④ What is Randomization Unit and Analysis Unit?

In any A/B experiment, two critical choices you must make are:

1. Randomization unit: The entity you randomly assign to control or treatment.
2. Analysis Unit: The entity you use to compute metrics for decision-making.

Simply,

- Randomization Unit: Who gets the treatment/control?
- Analysis Unit: Whose behavior do we measure to judge success?

② Real-world Analogy: Restaurant Experiment

Let's say a restaurant want to test if a new menu design increases revenue.

- They randomly assign restaurants (branches) to either show the old or new menu.
- But they measures revenue per customer.

So!

- Randomization Unit = restaurant (branches assigned treatment)
- Analysis Unit = customers (metric = revenue per customer).

This is not always 1:1. Sometimes we randomize on a group (cluster) but analyze on individuals.

Violation of independent observation.

Analogy: School Lunch Program.

A government wants to test whether a new lunch program improves student performances

They randomly assign entire schools to control or treatment - some get the new lunch program, some don't

- Randomization Unit = school.
- Analysis Unit = students.

Each student takes exams → their scores are the metrics. But we didn't randomise students, we randomised schools.

* why This's a problem:
Itamar what happens when:
- You randomize at group level (schools).
- But analyse at individual level (students)

You cannot assume one student are independent - students within a school are similar (same teachers, environment).

* So the core assumption of independent observation is violated.

* Math Under the hood
if you ignore this issue and run a simple t-test or regression at the student level:

$$Y_{ij} = \alpha + \beta \cdot Treatment_j + \varepsilon_{ij}$$

You're pretending each ε_{ij} is independent, but that's false. Student in the same school share:
- Noise factors (like school culture)
- Local policies
- Teacher quality.

This causes correlated errors ⇒ you underestimate standard errors ⇒ False positives (p-values too optimistic)

☑ Correct Approach: clustered Standard Errors.

Instead of assuming every student is independent, you tell your model

| "Treat student from the same school as a cluster".

That way, it accounts for intra-cluster correlation (ICC).

(3) Math Behind it (Simplified)

Let's denote:

- Y_{ij}: outcome for user j in group i
- $T_i \in \{0, 1\}$: treatment assigned at group level

We can estimate:

$$ATE = E[Y \mid T=1] - E[Y \mid T=0]$$

But if treatment is randomized at group level and analysis is at user level, we must account for intra-group correlation:

- Users within the same group are not independent.
- This inflates variance → needs clustered standard errors.

(4) Interlocking Topic: Unit of Diversion (from earlier chapters)

This relates to how users are bucketed into A/B groups!

↳ if diversion unit ≠ analysis unit, you must treat intra-cluster correlation carefully.
↳ similar trade-offs are discussed in Unit 5 and unit 13 ("Instrumentation")

→ clustered Randomization (unit 15)
→ Experiment Design (unit 5)
→ Statistical Analysis (unit 7, 8).
→ Instrumentation Design (unit 13)

(5) Summary

Term	Definitions	Example
Randomization Unit	Entity assigned to A/B group	store, user, session
Analysis Unit	Entity used to compute metrics	user, pageview.
Mismatch Risks	Biased or invalid inference	Must adjust for clustering.
Best Practice	Keep both unit aligned OR use cluster-robust stats.	

② User-level Randomization / User-level Randomization

① What is User-level Randomization?

It means:

> Each user is randomly assigned to variant (A or B), and they always see the same variant across sessions, devices, browsers, etc.

This is the most common and preferred randomization strategy in online experiments.

Why user-level?

Because users are:

- The core decision-maker (they click, buy, sign up, etc.)
- Typically independent of each other (unlike cookies, sessions, device)
- Easy to identify (via login, ID, or stable cookie)

This ensures:

- Clean assignment
- Repeated exposure consistency
- Statistically valid metrics.

② Real-world Analogy: Grocery Store Loyalty Cards.

You give every shopper a loyalty card:

- People with even card numbers → see Layout A
- People with odd card numbers → see Layout B.

Whenever a shopper comes back — even a week later — they still see their assigned layout.

This avoids pollution (where someone sees A today and B tomorrow — confusing and statistically invalid)

(3) Advantage of User-level Randomization

Benefits	Explanation
Consistency	same user always gets the same variant.
cleaner metrics	Conversion, retention, click-through all align with user
less noise	Reduce variance vs session / device randomization.
session stitching	Across device / browsers, still same variant (if ID is stable)

(4) When it can be challenging

~~If users:~~

If users:

- Don't log in
- Use multiple device without sync
- Are behind shared IP (e.g. cyber cafe is)

Then identifying unique is harder. Alternative (like cookie-level or session-level randomization) may be used - but they're less ideal.

(5) Links with Other Topics

Related Concept	Link.
Randomization Unit vs. Analysis Unit (p. 168)	→ In user-level randomization, both units usually align = ideal.
Cookie-level Randomization (unit 15)	→ Easier but noisy - a fallback when user-ID is unavailable.
Interaction (Unit 13)	→ Needs logging at user level, not just page session.
Tracking Pollution	→ Covered cookies in Twyman's Law and Logging chapters.

(6) Summary

Concept	Meaning
user-level Randomization	Assign each unique user (via ID or cookie) to A/B
why it's best	stable identity, clean exposure, valid metrics
use when	You can reliably track users (login, account, long-lived cookies)
Avoid when	No. stable user identity - fallback: session or cookie randomization.

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Best paired with Analysis at user-level, logging user-level metrics.

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